

Article

Experimental and Theoretical Study of Forced Synchronization of Self-Oscillations in Liquid Ferrocolloid Membranes

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Abstract: A ferrocolloid is a suspension of nanometer-sized ferromagnetic particles (magnetite) in a carrier liquid (kerosene). A unique feature of a ferrocolloid is the fact that layers consisting of densely packed particles are formed near the electrode surface under the influence of an external electric field. Each layer is a liquid membrane, and its formation significantly affects the various properties of the system. For example, the development of a unique phenomenon in a ferrocolloid is self-organization (self-oscillations and autowaves). The applied external periodic force leads to a change (capture) of the frequency of the autowave process—forced synchronization of autowaves. The experimentally obtained synchronization was investigated by the method of electrically controlled interference. After multiple experiments and theoretical studies, a physical mechanism for the synchronization of the autowave process in a cell with a ferrocolloid was proposed for the first time. A mathematical model of forced synchronization of autowaves, which is described by a system of nonlinear differential equations, was proposed for the first time as well. Adding an external periodic force into the model led to a change in the frequency of autowaves; synchronization by an external force was confirmed by computational experiments.

Keywords: ferrocolloid; membranes; mathematical modeling; nonlinear processes; synchronization



Citation: Chekanov, V.; Kovalenko, A.; Kandaurova, N. Experimental and Theoretical Study of Forced Synchronization of Self-Oscillations in Liquid Ferrocolloid Membranes. *Coatings* **2022**, *12*, 1901. <https://doi.org/10.3390/coatings12121901>

Academic Editor: Jinyang Xu

Received: 28 September 2022

Accepted: 30 November 2022

Published: 5 December 2022

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1. Introduction

Liquid-dispersed nanostructured systems (ferrocolloids or magnetic fluids) have unique electrical and electro-optical properties, which are of scientific interest from the point of view of studying the interaction of dispersed particles with external fields and with each other [1–5]. A ferrocolloid is a colloidal system of ferromagnetic particles (magnetite) with a size of ~10 nm in a carrier liquid (kerosene). To avoid coagulation, particles are usually covered with a protective shell of oleic acid [6]. A unique feature of ferrocolloids is the formation of dense layers of magnetite particles with a protective shell near the electrodes in a constant electric field. The layer does not mix with the main volume of the liquid and behaves as a liquid ferrocolloid membrane [7–10]. The membrane is formed as a result of electrophoresis of particles of the dispersed phase. It is a stable structure, the thickness of which depends on the magnitude of the applied electric field. The stability of the membrane comes from the magnetic interaction between magnetite particles [11]. The formation of a liquid membrane has a significant effect on the macroscopic properties of a ferrocolloid. The liquid ferrocolloid membrane has a special property—the recharging of colloidal particles [12]. In turn, the recharging of particles in the membrane is the basis of the physical mechanism for the development of self-sustaining nonlinear waves (autowaves) [13].

Autowaves—one of the examples of self-organization of matter—are currently the object of close attention for specialists in various scientific fields [14–17].

Interest in this topic is caused primarily by the fact that the laws of behavior and propagation of autowaves are the same and are used to explain phenomena in biology,

chemistry, medicine, urban studies, and other fields. Autowaves in the near-surface layer of a ferrocolloid (magnetic fluid) are a unique phenomenon. The ease of reproduction makes the autowave process (AWP) convenient for research. The number of cycles flowing in it can exceed several thousand. Therefore, a cell with a ferrocolloid is a good model environment for studying and simulating ARP in biological objects, such as a heart muscle or a nerve fiber. Autowaves in a ferrocolloid were studied in [13,18,19].

One of the main trends in the living world is the tendency to achieve a common rhythm in collective behavior—the tendency to synchronize.

The term “synchronization” refers to a wide class of phenomena from the scientific and social fields. Adjustment of the heart rate to breathing, birds in a flock, violins in an orchestra, and the radiation of a population of fireflies are phenomena that seem different but obey the same laws: the object adjusts the rhythm of its actions to other objects. Resonance interactions leading to the observation of standing waves have previously been observed in biological systems (e.g., circadian rhythms).

The main provisions of the theory of synchronization are most fully described in monographs [20–23]. It is known [20] that if a self-oscillatory system is affected by an external periodic force, the frequency of which is a multiple of the natural frequency of self-oscillations, then synchronization (or capture) of the frequency occurs, that is, the frequency of self-oscillations becomes exactly the same number of times less or more than the frequency of the external force. If the frequency of the external force is close to the frequency of its own self-oscillations, one speaks of synchronization on the fundamental tone.

Synchronization of autowaves in chemical systems (the Belousov-Zhabotinsky reaction) was obtained in [24], wherein it was experimentally shown that the reaction–diffusion processes that rule the evolution and selection of patterns in many chemical and biological systems can also exhibit frequency-blocking phenomena.

In the present work, forced synchronization (frequency blocking) of an autowave process in a thin layer of a ferrocolloid by an external periodic electric force was experimentally obtained for the first time for different ratios of the natural oscillation frequencies and the frequency of the driving force. A physical synchronization mechanism was established. A mathematical model of synchronization of the autowave process, which takes into account the action of the driving force, was proposed.

2. Materials and Methods: Experimental Technique

The experimental medium was a thin layer of ferrocolloid (ferrocolloid membrane) placed between two electrodes. One of the electrodes was glass with a conductive transparent ITO (indium tin oxide) coating, a semiconductor material that is transparent to visible light (Figure 1a). A glass prism was installed on the electrode to eliminate the flare effect.

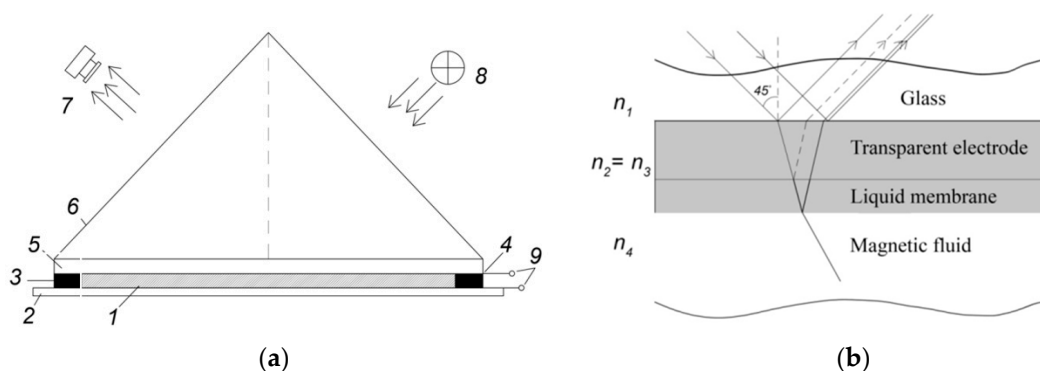


Figure 1. (a) Schematic of an experimental device: 1—ferrocolloid; 2 and 5—ITO-covered glass; 3 and 4—polystyrene insulating gaskets; 6—rectangular isosceles prism; 7—camera; 8—light source; 9—electrodes. (b) Multilayer structure: glass–transparent electrode–liquid membrane–ferrocolloid. The light path is shown by arrows.

The concentration of the dispersed phase in the liquid used in the experiment was 9 vol.%. The thickness l of the ferrocolloid layer in the cell was $280\text{ }\mu\text{m}$. The surface area of the electrodes was $S = 36 \times 30\text{ mm}^2$. In the experiments, we used two samples of ITO-coated glasses (manufactured by LLC Politech, St. Petersburg). The glass thickness was 4 mm, and the thickness of the conductive coating was $h_{01} = (280 \pm 5)\text{ nm}$. The measurements were taken at five different points on the surface and then averaged. The optical parameters of the ITO coating were measured with an SE 800 SENTECH spectroscopic ellipsometer. According to measurements at a wavelength of $\lambda = 650\text{ nm}$ at ten points at a distance of 5 mm from each other, after averaging, the complex refractive index of the conductive coating was $n_1 = 1.76 (1 + 0.04i)$. The refractive index of glass was $n_2 = 1.52$, and the refractive indices of the ferrocolloids with concentrations of 9 vol.% and 27 vol.% were $n_3 = 1.48 (1 + 0.01i)$ and $n_4 = 1.75 (1 + 0.03i)$, respectively.

When a cell with a ferrocolloid was placed in a constant electric field, a liquid membrane was formed at the electrodes, the thickness of which increased with increasing voltage. If the surface of the cell was illuminated with white light, then the change in the membrane thickness was visually observed with the changing of the surface color. This effect was associated with the interference of falling rays from the upper and lower boundaries of the liquid membrane (Figure 1b) and this is called electrically controlled interference [25]. After the voltage on the electrodes increased to the value V_{cr} , the membrane became unstable, and the concentration of magnetic particles fluctuated with a frequency ω_0 (autowave process (AWP)) (Figure 4a). The liquid membrane is a reaction-diffusion medium and can be considered as a two-dimensional array of nonlinear oscillators [26–28]. To determine the oscillation frequency ω_0 , a cell with a ferrocolloid was illuminated with monochromatic light (a laser with a wavelength of $\lambda = 650\text{ nm}$) at an angle of 45 degrees (Figure 2).

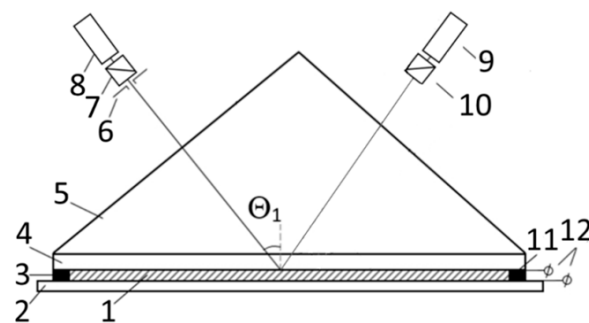


Figure 2. Experimental setup for determining the frequency of fluctuations in the concentration of particles in a liquid membrane. 1—ferrocolloid; 2—a plate of foil getinax; 3 and 11—insulating gaskets; 4—ITO-covered glass; 5—rectangular isosceles prism; 6—diaphragm; 7 and 10—Polaroid camera; 8—photodiode; 9—laser; 12—electrodes.

The beam reflected from the glass-conductive coating and conductive coating-liquid membrane surfaces and continued to the PD-256 photodiode through a diaphragm (6 in Figure 2) and a Polaroid camera (7 in Figure 2). The photodiode was connected to the GDS-71022 oscilloscope, which showed the dependence of the voltage on the photodiode on time (Figure 3, beam 2). Taking into account that the photodiode operated in a linear mode, the voltage at the input resistor of the oscilloscope U_{osc} was proportional to the reflected light intensity and proportional to the coefficient reflections R : $U_{osc}(t) = kR(t)$. The shape of the curves in Figure 3 is explained as follows: at a certain critical voltage V_{cr} ($\sim 8\text{ V}$), the intensity of the reflected light began to change periodically (beam 2), which indicated a periodic change (fluctuations) in the layer thickness. The moment of occurrence of self-oscillations was considered to be the voltage V_{cr} at which the oscillogram of the optical response took a periodic form. The period (frequency) of oscillations was taken from the oscillogram. Beam 1 is the ampere-time characteristic of the process.

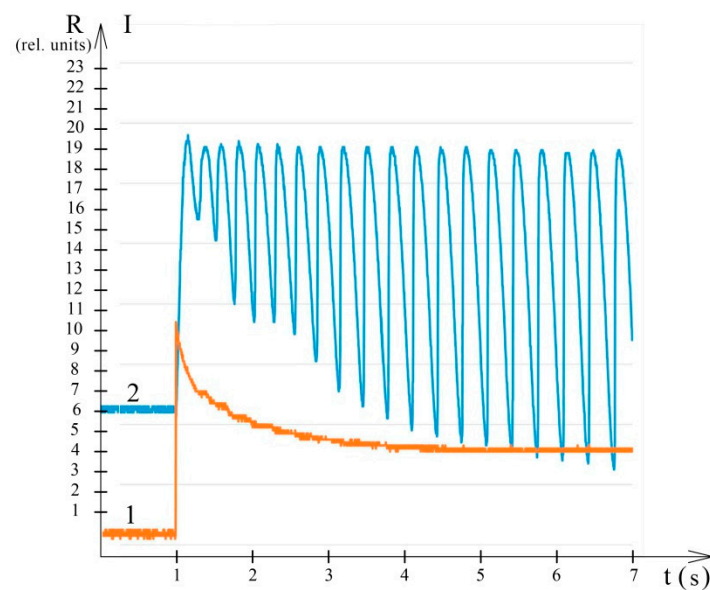


Figure 3. Plotted graph based on the oscillogram obtained by the GDS-71022 oscilloscope. Dependence of the reflection coefficient of the cell surface on time (optical response) during self-oscillations of the particle concentration in the liquid membrane. Beam 1—current passing through the cell; beam 2—periodic change in the intensity of reflected light (optical response).

Simultaneously, with a constant voltage V , a cell with a ferrocolloid was affected by a rectangular pulsed voltage U , the amplitude of which was less than the amplitude of the constant voltage, so it did not suppress autowaves.

The frequency of the external force was varied in such a way as to obtain zero detuning or a very small one ($\omega_0 - \omega_p$). With a change in the pulse frequency ω_p from 0.9 to 5 Hz, after some time t , the frequency was captured, and a sequence of synchronization patterns shown in Figure 4b–e was observed. Oscillations had the observing frequency Ω , which in a private case of zero detuning was equal to the natural frequency ω_0 . Video files are available at <https://disk.yandex.ru/d/ku3tiAlaITWjpQ>, accessed on 4 December 2022.

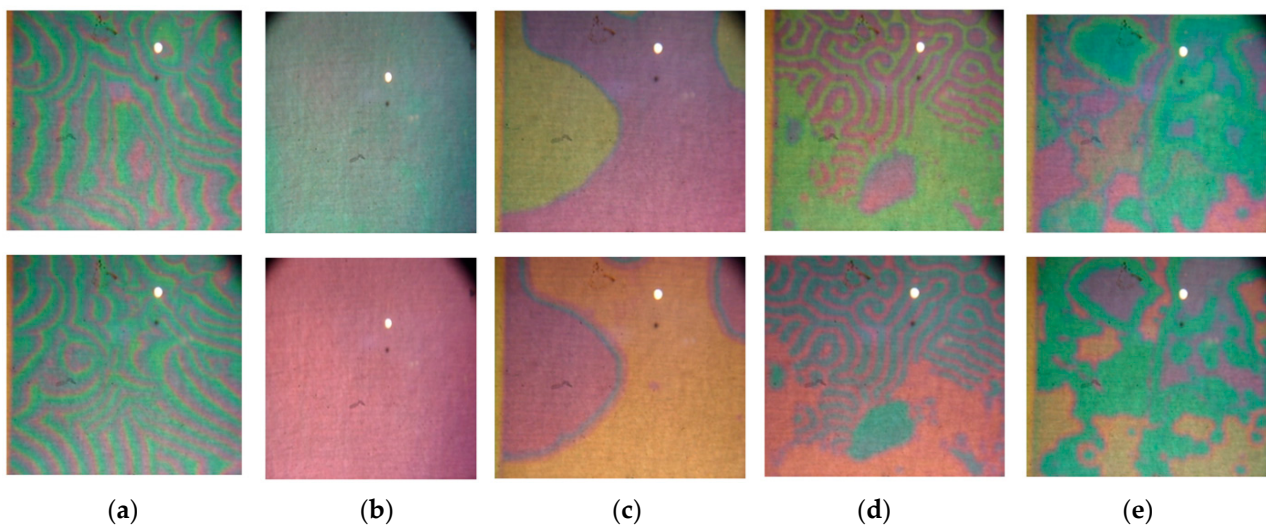


Figure 4. Illustration of various frequency-blocked modes of the AWP in a cell under the influence of a periodic electric force (pulse voltage). Frequency ratios: (a) 0, (b) 1:1, (c) 1:2, (d) 1:2.5, and (e) 1:3. The exposure times in the upper and lower rows differ by half of a period.

At constant voltage $V = 8.1$ V, impulse voltage $U = 2.5$ V, and frequency $\omega_p = 1.35$ Hz, the entire surface of the cell with the ferrocolloid changed color (which meant that the

thickness of the entire membrane changed). This means that the detuning was zero and the frequency of concentration oscillations of the entire cell surface was equal to the frequency of the external periodic force— $\omega_p:\omega_0 = 1:1$ (Figure 4b). Figure 5 shows the time dependence of the intensity of the reflected light (beam 1) and the type of impulse voltage (beam 2), which shows that the period of concentration fluctuations was equal to the period of the impulse voltage, so the frequency capture occurred.

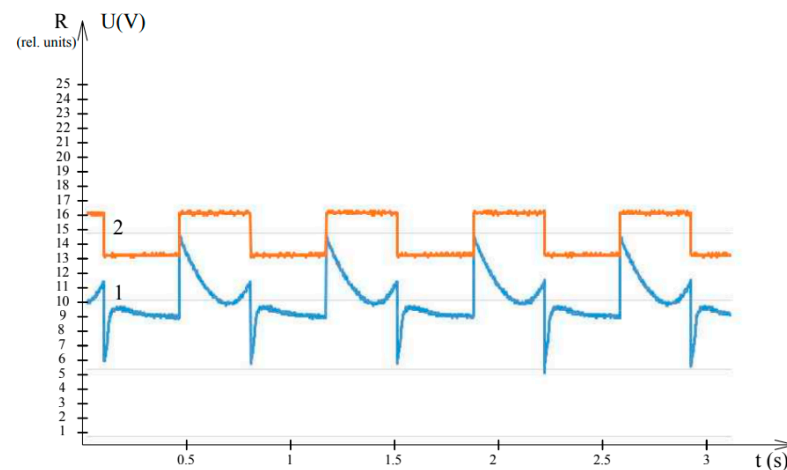


Figure 5. Plotted graph based on the oscillogram obtained by the GDS-71022 oscilloscope. Forced synchronization of fluctuations in the concentration of the dispersed phase of the ferrocolloid. Beam 1—optical signal: time dependence of the intensity of reflected light; beam 2—pulse voltage $U = 2.5$ V, frequency $\omega_p = 1.35$ Hz.

Figure 4c shows the synchronization of oscillations at the frequency ratio $\omega_p:\omega_0 = 2:1$. The constant voltage on the electrodes was $V = 16$ V, and the amplitude of the pulse voltage was $U = 8$ V. It can be seen that two spatially homogeneous regions of the liquid membrane oscillated in antiphase. The strip separating these regions (in Figure 4c it is blue) was a stationary front, its color (and hence the membrane thickness in this region) did not change, and there were no concentration fluctuations. In [24], such strips are called “nodal lines”. The synchronization modes listed above can go over many cycles.

With a frequency ratio $\omega_p:\omega_0 = 1:2.5$, standing waves or labyrinth structures were obtained (Figure 4d).

The experimental obtaining of synchronization of reverberators—spiral waves or autowave vortices—deserves special attention. A reverberator arose as a result of the evolution of a flat autowave front break. Reverberators rotated at a variable speed, so the formation of standing waves in the event of periodic external force acting on the reverberator was impossible. In the experimental cell in the ferrocolloid, the modulation of the reverberator speed was obtained (Figure 6).

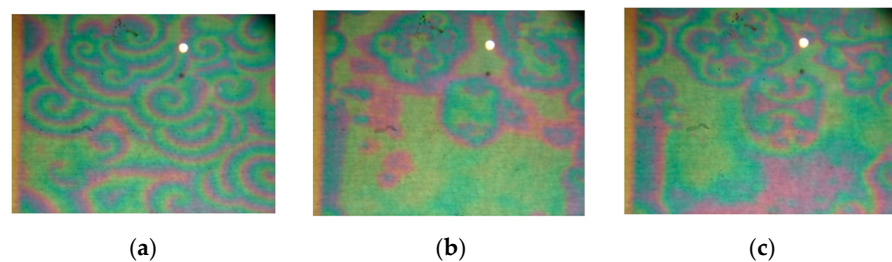


Figure 6. Modulation of the reverberator rotation frequency under the influence of a periodic external force: (a) autowave process without external periodic influence and (b,c) rotation of the reverberator under additional external periodic action.

Video files are available at <https://disk.yandex.ru/d/ku3tiAlaITWjpQ> accessed on 4 December 2022.

3. Discussion of Experimental Results

It is known [20] that if a self-oscillating system, which oscillates with a natural frequency ω_0 , is acted upon by an external periodic force, the frequency of which is ω_p , and if the detuning $(\omega_0 - \omega_p)$ is sufficiently small, then the frequency of the observed oscillations Ω becomes equal to the frequency of the external force—this is called synchronization (or frequency capture). The coincidence of frequencies in a finite range of detuning is the main property of synchronization. In a more complicated case, self-oscillations with a natural frequency ω_0 can be captured by a force with a frequency close (but not necessarily equal) to $\omega_0/2$, $\omega_0/3$, etc. Then the frequency of the observed oscillations will be equal to $\Omega = 2\omega_p$ or $\Omega = 3\omega_p$. This mode is called $n:m$ synchronization. Since we consider the autowave process in the near-electrode layer of a ferrocolloid from the position of the oscillations of coupled oscillators [29,30], each element of the layer (particle ensemble) interacts with its nearest neighbors or, in a more complex case, with several neighbors. Each oscillator has its own frequency, but there may also be clusters of oscillators that oscillate with the same frequency [31].

Sections of the near-electrode layer can be considered as a system of coupled nonlinear oscillators, each of which tends to synchronize its motion with the neighboring one and with an external harmonic force. If the oscillators do not interact with each other, then with the frequency ratio $\omega_0:\omega_p = 1:m$, where m is an integer, neighboring oscillators would have equal phase differences $2\pi/m \cdot i$, where $i = 1, \dots, m-1$. However, since the oscillators are coupled (including due to the magnetic dipole–dipole interaction of ferrocolloid particles), they tend to have the same phases as their neighbors. Clusters of particles oscillate with the same frequency, and clouds are formed. The phase difference between the clouds is $2\pi/m \cdot i$ [20].

The synchronization mode shown in Figure 4c, when the frequency of the observed oscillations Ω and the frequency of the driving force ω_p are equal, demonstrates a simultaneous periodic change in the color of the entire surface of the cell with a ferrocolloid—the entire medium oscillates uniformly. This mode is observed when the natural oscillation frequency ω_0 is equal or close to the frequency of the driving force.

When we changed the frequency of a periodic force (impulse voltage) in such a way that two regions were formed on the cell surface that oscillated in antiphase (Figure 4c), the frequency of the observed oscillations Ω and the frequency of the driving force were related as $\omega_p:\Omega = 1:2$. The natural frequency ω_0 will be close or equal to Ω . At this frequency ratio, a cell with a ferrocolloid acted as a bistable system with two states corresponding to two possible vibrational phases. It is very interesting that between two boundaries that oscillate in antiphase there was always a stationary region that did not oscillate. This result was also observed in [24,32], which supports the theoretical hypothesis that such behavior is a fundamental property of interfaces in reliable systems.

An interesting result was obtained with a frequency ratio $\omega_p:\Omega = 1:2.5$ (Figure 4d); the photograph shows labyrinth structures—standing waves. In [24], similar structures appeared only when the intensity of the external action was above a certain critical value. In our experiments, the appearance of labyrinths depended only on the ratio of the frequencies of the external force and the natural oscillation frequency. The amplitude of the external force in all experiments was less than the amplitude of the constant force, which is the source of energy for the generation of autowaves.

4. Mathematical Model of Forced Synchronization of an Autowave Process in a Ferrocolloid

A mathematical model was developed that confirms the synchronization of the AWP in a ferrocolloid by a periodic electric force. It is based on the model of a self-oscillatory process in a ferrocolloid, which is described in [33]. The model consisted of a research

area (Figure 7), a system of Equations (1)–(7) and initial and boundary conditions in Equations (8)–(15).

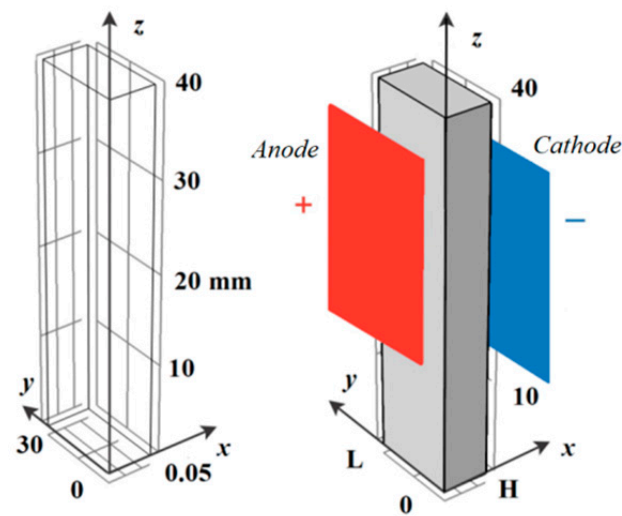


Figure 7. Scheme of a 3D cell with a magnetic fluid (electrodes are highlighted in color) with research areas in the form of a rectangular parallelepiped.

Let $x = 0$ and $x = H$ be the conditional points at which the process of charging particles takes place (no space charge in $x = 0$); in a particular case, these can be an anode and a cathode. The initial distribution of neutral particles of a ferrocolloid with a known concentration (initial conditions $C_1(0, x) = C_{10}(x)$ and $C_2(0, x) = C_{20}(x)$) is determined by diffusion. In an electric field at the initial moment of time $t = 0$, the particles become charged. Around the point $x = H$, the particles become charged negatively, and around $x = 0$, they become charged positively. In an electric field, negatively charged magnetic particles begin to move towards the anode, and positively charged ones move towards the cathode, so the dense layers form near the electrodes.

The model of AWP is based on the mechanism of particle recharging in layers and is a classical system of differential equations [34–37], without simplifications and fitting parameters:

$$\vec{j}_i = -\frac{F}{RT} z_i D_i C_i \vec{E} - D_i \nabla C_i + C_i \vec{V}, \quad i = 1, 2, \quad (1)$$

$$\frac{\partial C_i}{\partial t} = -\text{div} \vec{j}_i, \quad i = 1, 2, \quad (2)$$

$$\varepsilon_r \Delta \varphi = -F(z_1 C_1 + z_2 C_2), \quad (3)$$

$$\vec{I} = F(z_1 \vec{j}_1 + z_2 \vec{j}_2) \quad (4)$$

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \nabla) \vec{V} = -\frac{1}{\rho_0} \nabla P + \nu \Delta \vec{V} + \frac{1}{\rho_0} \vec{f}, \quad (5)$$

$$\text{div}(\vec{V}) = 0, \quad (6)$$

$$\vec{f} = \rho \vec{E} \quad (7)$$

The Nernst–Planck Equation (1) describes the charged particles' flow due to diffusion convection and electric migration, the charge numbers of positively charged particles $z_1 = 1$, and negatively charged particles $z_2 = -1$. Equation (2) is the material balance equation, Equation (3) is the potential Poisson equation, and Equation (4) is the current flow equation, where ε_r is the dielectric constant, φ is the potential, C_i is the concentration, j_i is the flux, D_i is the diffusion coefficient, $\vec{I}$ is the current density, and $\vec{V}$ is the flow rate

of the magnetic fluid. The Navier–Stokes Equation (5) and continuity equations for an incompressible fluid (6) describe the velocity field formed, in particular, under the action of a spatial electric force (7), with $\vec{f}$ being electric force density, where $\rho = F(z_1C_1 + z_2C_2)$ is the space-charge density, ρ_0 is the ferrocolloid density, and ν is the kinematic viscosity.

The process of the wave appearance is associated with the process of charging and recharging of the particles, which is reflected in the boundary conditions in Equations (8)–(15).

At all boundaries of the study area, the adhesion condition is used for the velocity. The initial condition is defined as a stable solution.

$$V_x(0, x, y, z) = V_y(0, x, y, z) = 0. \quad (8)$$

Voltage on the electrodes is higher than zero.

To describe the recharge process of MF particles in a liquid membrane, we used the following boundary conditions:

At the anode ($x = 0$):

$$\left(-\frac{F}{RT_0} D_1 z_1 C_2 \frac{\partial \varphi}{\partial x} - D_1 \frac{\partial C_2}{\partial x} \right) \Big|_{x=0} = j_{1A} \quad (9)$$

$$\left(-\frac{F}{RT_0} D_2 z_2 C_1 \frac{\partial \varphi}{\partial x} - D_2 \frac{\partial C_1}{\partial x} \right) \Big|_{x=0} = j_{2A} \quad (10)$$

At the cathode ($x = H$):

$$\left(-\frac{F}{RT_0} D_1 z_1 C_2 \frac{\partial \varphi}{\partial x} - D_1 \frac{\partial C_2}{\partial x} \right) \Big|_{x=H} = j_{1K} \quad (11)$$

$$\left(-\frac{F}{RT_0} D_2 z_2 C_1 \frac{\partial \varphi}{\partial x} - D_2 \frac{\partial C_1}{\partial x} \right) \Big|_{x=H} = j_{2K} \quad (12)$$

The recharge conditions in Equations (9)–(12) reflect the process of two flows (positive and negative) turning into each other after recharging on both electrodes.

The model considers the equipotentiality of the cathode and anode surfaces as the boundary conditions for the potential

$$\begin{aligned} \varphi(t, 0, y, z) &= \alpha \\ \varphi(t, H, y, z) &= 0 \end{aligned} \quad (13)$$

Insulators are considered to be impermeable:

$$-\vec{n} \cdot \vec{j}_i = 0, \quad i = 1, 2 \quad (14)$$

$$-\vec{n} \cdot \nabla \varphi = 0 \quad (15)$$

At the initial moment of time, the particles are concentrated on both electrodes; i.e., $C_{10}(0, x, y, z)$, $C_{20}(0, x, y, z)$ are located near the anode $x = 0$ and cathode $x = H$.

For the initial distribution of magnetic particles, we used the following functions:

$$C_1(0, x, y, z) = C_{10}(0, x, y, z) = 0.074 \cdot e^{-x/(0.01 \cdot H)} \text{ mol/m}^3 \quad (16)$$

$$C_2(0, x, y, z) = C_{20}(0, x, y, z) = 0.074 \cdot e^{-(H-x)/(0.01 \cdot H)} \text{ mol/m}^3 \quad (17)$$

$$\varphi(0, x, y, z) = \alpha - \frac{\alpha x}{H} \quad (18)$$

To simulate AWP synchronization, we considered two different options for setting $\Delta\varphi = U$: in the first case, we took a fixed value of the voltage, for example, $\Delta\varphi = 4V$. It was shown in [13] that for a fixed voltage, self-oscillations arise in the system, the frequency of which depends on this jump. For example, with a voltage $\Delta\varphi = 4V$, the frequency of

natural oscillations is 18.5 Hz. In the second step, we set a periodic change in the voltage with a 25% amplitude relative to the initial one, so that the oscillation frequency is almost equal to but slightly different from the natural frequency of self-oscillations. For example, for the case $\Delta\varphi = 4V$, we took the forcing oscillations $2 \sin(10t)$. Thus, the total voltage has the form $\Delta\varphi = 4 + 2 \sin(10t)$.

5. Results of the Implementation of the Mathematical Model (Computer Experiment)

The numerical solution is based on the finite element method and the splitting of the problem at each current layer in time into the hydrodynamic (Navier-Stokes equation) and electrochemical (Nernst-Planck-Poisson) parts with their sequential solution to complete convergence of the specified accuracy. This method was successfully applied to solve some problems of physical chemistry [7,33,38,39].

The method was implemented in the software developed by the authors, "1D modeling of an autowave process in a thin layer of a magnetic colloid (AutoWave01)" [40], created in the Comsol Multiphysics environment.

The convergence of the solution and the quality of the grid were checked by reducing the grid step by half, and the results coincided. The grid spacing decreased (the number of elements increased) until the calculation results stopped changing (the results coincided).

The parameters of the task were $H = 5 \cdot 10^{-5}$ m; the calculation was carried out for $t \in [0, 10]$ seconds with a memorization step of 0.01 s. The mesh parameters were $n = 100,000$ elements. In the calculations below, a certain initial distribution of the concentration of the magnetic parts corresponding to the experimental data is accepted:

$$C_1(0, x) = C_2(0, x) = 9 \text{ vol.}\%$$

The volumetric concentration of magnetite particles expressed in mol/m^3 is

$$C = 0.73593 \cdot 10^{-3} \text{ mol}/\text{m}^3$$

$$C_1(0, x) = C_{10}(x) = 0.0074 \cdot e^{-x/(0/01 \cdot H)} \text{ mol}/\text{m}^3$$

$$C_2(0, x) = C_{20}(x) = 0.0074 \cdot e^{-(H-x)/(0/01 \cdot H)} \text{ mol}/\text{m}^3$$

According to these initial conditions, initially, the particles were distributed near the electrodes quite densely, that is, the liquid membrane had already formed. The results of a numerical analysis for two waves were obtained.

As a result of the implementation of the computer experiment, the graphs shown in Figure 8a,b were obtained.

Figure 8a shows that over time, the natural oscillations of the system (green color) became adjusted to the forcing ones (red color), i.e., process synchronization started after 5 s, because at the beginning, the process was established, and then it was captured by an external influence.

Let us calculate the frequency of natural oscillations of the system; since the period of natural oscillations of the system $T = 0.34$ s, then $\omega_0 = 18.5$ Hz is the frequency of natural oscillations of the system. Let us take the frequency of external oscillations less than two times, i.e., $\omega_p = 9.25$ Hz.

From the analysis (Figure 8a,b) of the AWP synchronization by an external periodic action with different parameters, for example, $\sin(10t)$ and $\sin(40t)$, it can be seen that initially (up to 5 s), external periodic forcing oscillations practically did not affect the system. However, it is clearly seen that after 5 s (Figure 8a), the oscillations of the system became adjusted to the forcing ones with a frequency of external oscillations $\omega_p = 10$ Hz. If the driving oscillations had a frequency $\omega_p = 40$ Hz, then the adjustment of the system oscillations began a little later than 6 s (Figure 8b).

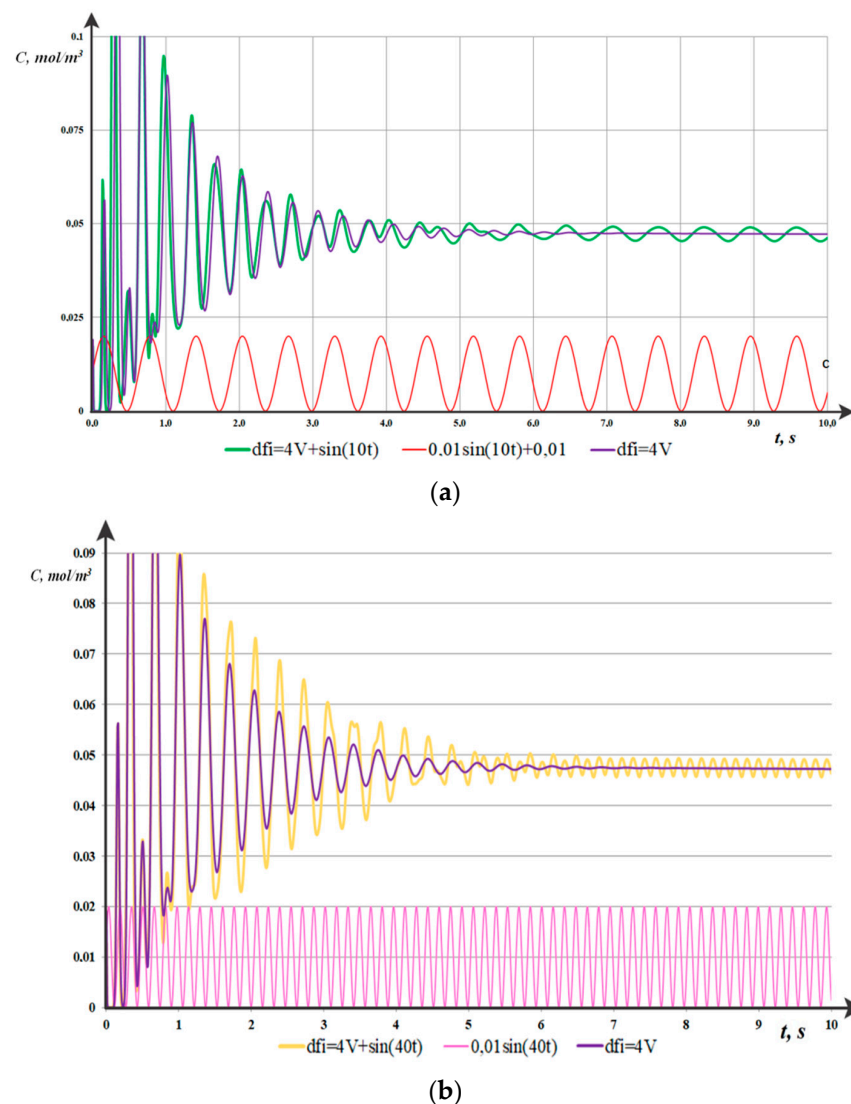


Figure 8. (a) Periodic change in the total concentration (self-oscillations) on the cathode surface at a potential jump of 4 V (purple), the same self-oscillations with forcing oscillations $\Delta\varphi = 4 + \sin(10t)$ (green) and periodic forcing oscillations $0.01 \sin(10t)$ (red). (b) Periodic change in the total concentration (self-oscillations) on the surface of the cathode with the potential jump of 4 V (purple), as well as the same oscillations with force fluctuations $\Delta\varphi = 4 + \sin(40t)$ (yellow) and periodic forcing vibrations $0.01 \sin(40t)$ (pink).

6. Conclusions

1. The action of an external periodic force on an autowave in a ferrocolloid led to a change in the frequency of observed autowaves. If the frequency detuning was small, then the natural frequency of the autowaves ω_0 became equal to the frequency of the external force ω_p or proportional to it.
2. The periodic application of electrical force transformed spiral waves. Since the reverberator rotated at a variable speed, the frequency of its rotation was modulated.
3. A sequence of frequency-blocked modes was observed by changing the frequency of external periodic exposure. The structures obtained experimentally in a cell with a ferrocolloid under the action of an external periodic force are similar to those described in [24], where the same structures were observed by the authors in a quasi-two-dimensional reaction-diffusion system with a light-sensitive Belousov-Zhabotinsky reaction. This result is convincing evidence that the phenomenon of synchronization

when pairing the internal oscillatory dynamics with an external periodic action can arise in a variety of chemical, biological, and other systems [41–46].

4. A mathematical model of the synchronization of self-oscillations was developed as a boundary value problem for a nonlinear system of partial differential equations, and a numerical solution was obtained. The frequency capture of autowaves by an external periodic force was confirmed in a computer experiment, which shows the adequacy of the developed model. Mathematical modeling of autowave synchronization confirmed the fact that a complex of coupled nonlinear oscillators can exhibit spatial reorganization under the influence of an external periodic action.

Author Contributions: Conceptualization and methodology, V.C. and A.K.; software and writing—original draft preparation, V.C. and A.K.; resources and data curation, N.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Acknowledgments: This research was carried out within the framework of the work of the International Laboratory “Computer and mathematical modeling of nonlinear processes”.

Conflicts of Interest: The authors declare no conflict of interest.

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